

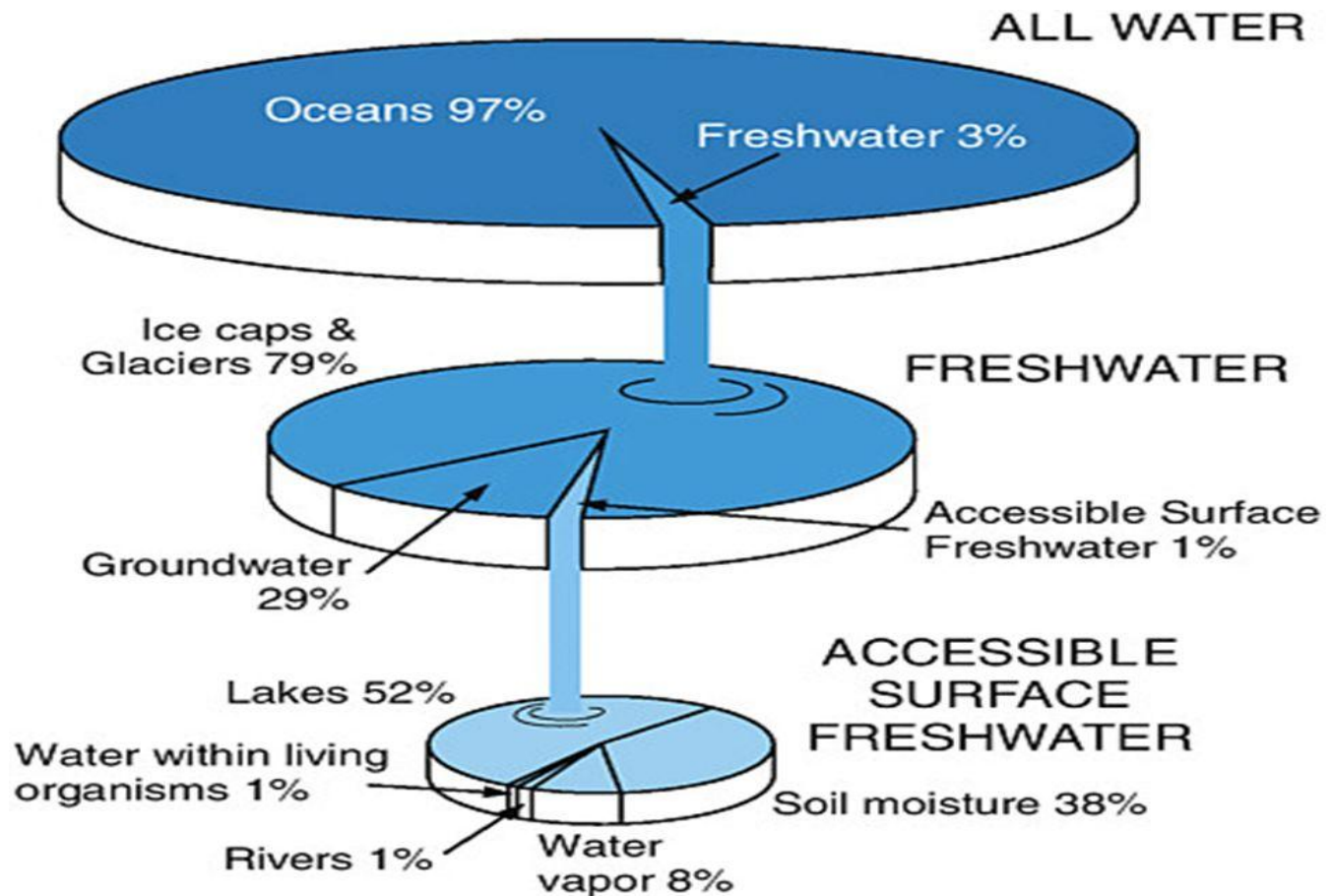
Water Resource





Water Resource

Distribution of Global Water





Water Resource

Hydrological Cycle.

The water cycle, also known as the hydrologic cycle or the hydrological cycle, describes the continuous movement of water on, above, and below the surface of the Earth.

A fundamental characteristic of the hydrologic cycle is that it has no beginning, and it has no end. It can be studied by starting with any process: evaporation, condensation, precipitation, infiltration, transpiration, runoff, and storage.

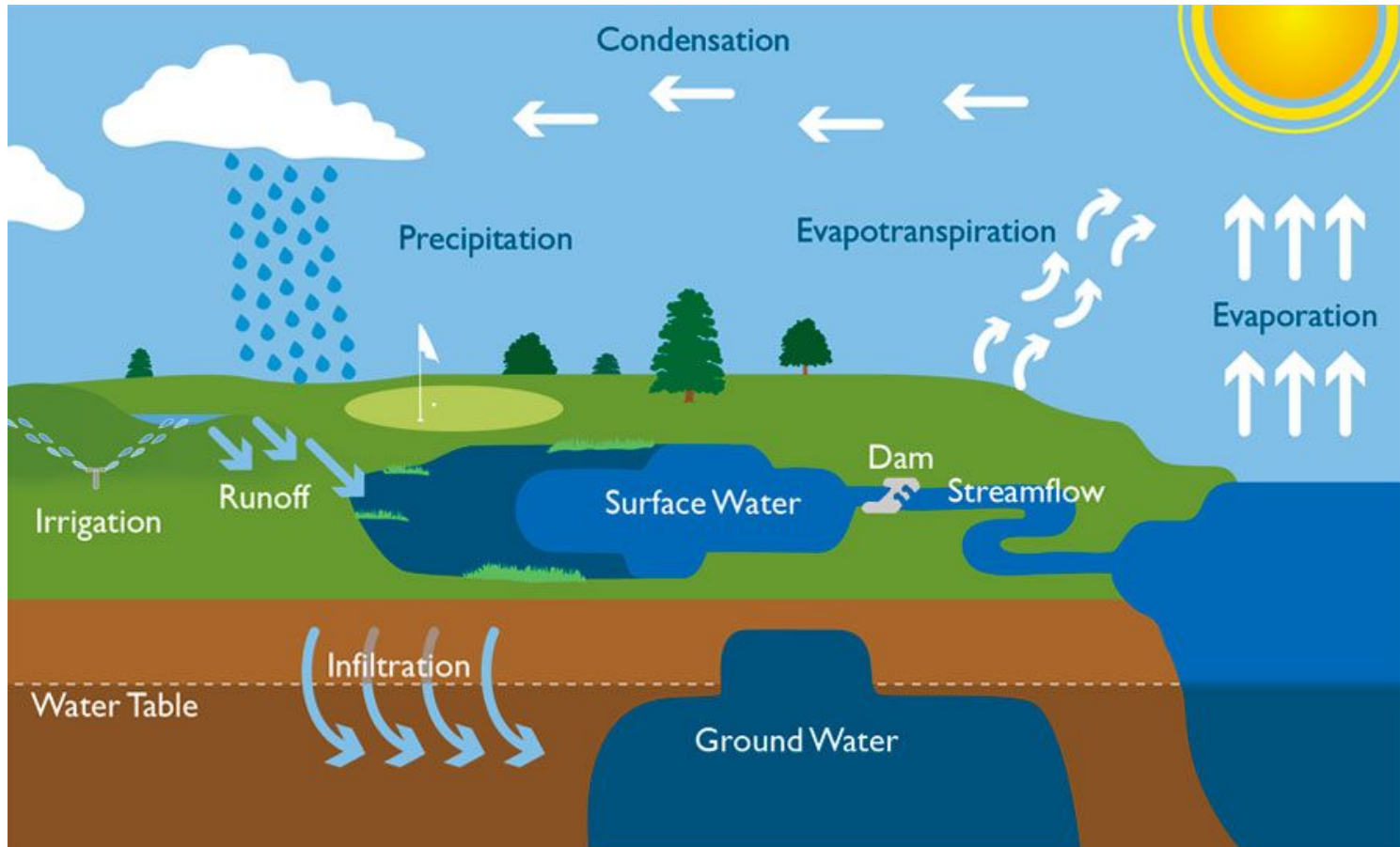
Evaporation occurs when water's physical state changes from a liquid to a gaseous state. During this change, a considerable amount of heat, about 600 calories of energy for each gram of water, is exchanged. Typically, solar radiation and other factors such as air temperature, vapor pressure, wind, and atmospheric pressure affect the amount of natural evaporation in any geographic area.

Condensation is the process by which water vapor changes its physical state from a vapor, most commonly, to a liquid. Water vapor condenses onto small airborne particles to form dew, fog, or clouds. The most active particles that form clouds are sea salts, atmospheric ions caused by lightning, and combustion products containing sulfurous and nitrous acids. Condensation is brought about by cooling the air or increasing the vapor in the air to its saturation point.



Water Resource

Hydrological Cycle Process



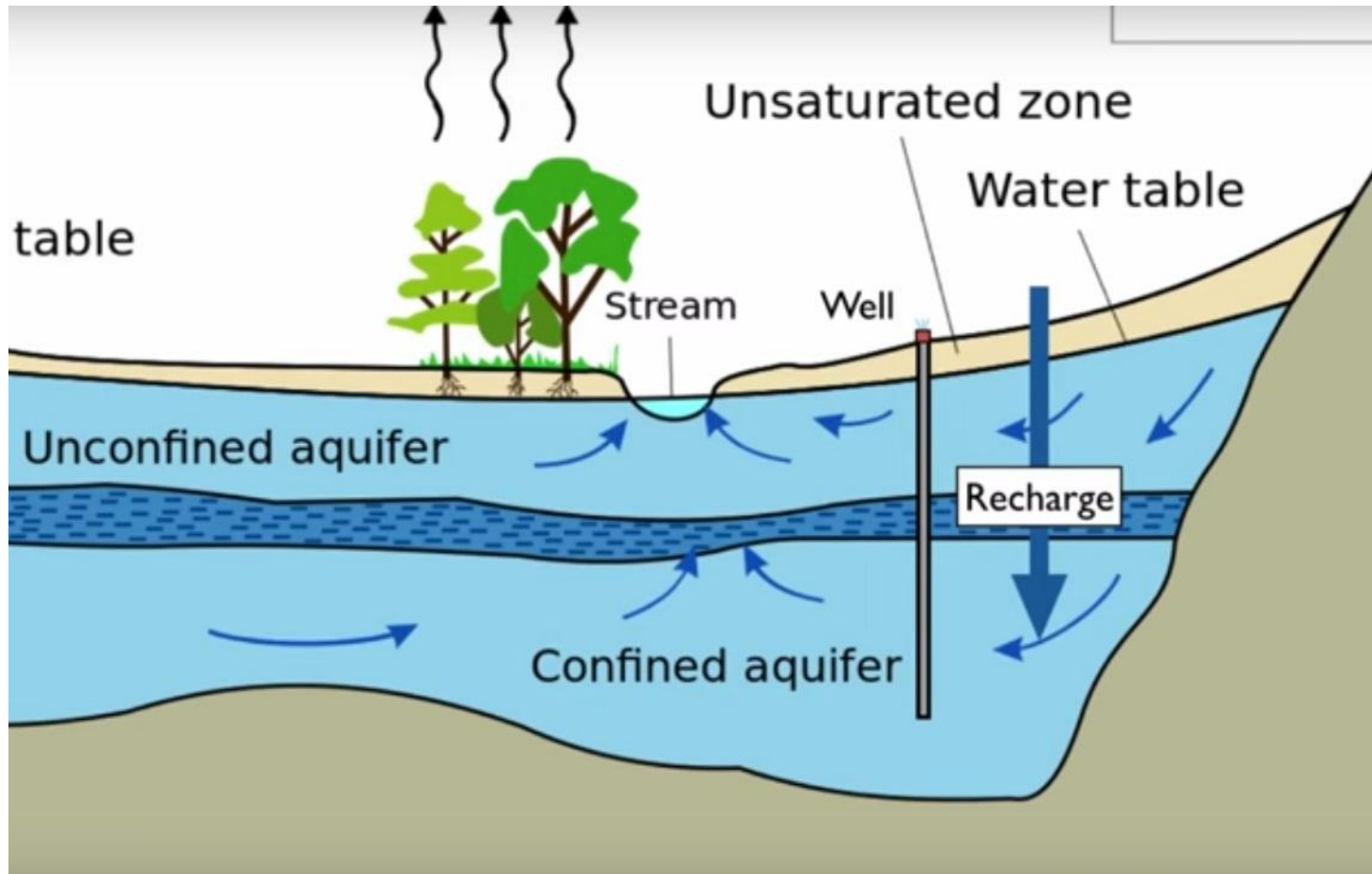
- **Precipitation** is the process that occurs when any form of water particles fall from the atmosphere and reach the ground. Two sub-processes cause clouds to release precipitation, the coalescence process and the ice-crystal process. As water drops reach a critical size, the drop is exposed to gravity and frictional drag. The other sub-process that can occur is the ice-crystal formation process. It occurs when ice develops in cold clouds or in cloud formations high in the atmosphere where freezing temperatures occur. When nearby water droplets approach the crystals, some droplets evaporate and condense on the crystals. The crystals grow to a critical size and drop as snow or ice pellets. Sometimes, as the pellets fall through lower-elevation air, they melt and change into raindrops. The portion of precipitation that appears in surface streams is called **runoff**.
- **Infiltration** is the physical process involving the movement of water through the boundary area where the atmosphere interfaces with the soil. The surface phenomenon is governed by soil surface conditions. Water transfer is related to the porosity of the soil and the permeability of the soil profile. Typically, the infiltration rate depends on the puddling of the water at the soil surface by the impact of raindrops, the texture and structure of the soil, the initial soil moisture content, the decreasing water concentration as the water moves deeper into the soil filling of the pores in the soil matrices, changes in the soil composition, and to the swelling of the wetted soils that in turn close cracks in the soil.
- Water that is infiltrated and stored in the soil can also become the water that later **evapotranspired or becomes subsurface runoff**.

- **Transpiration** is the biological process that occurs mostly in the day. Water inside of plants is transferred from the plant to the atmosphere as water vapor through numerous individual leaf openings. Plants transpire to move nutrients to the upper portion of the plants and to cool the leaves exposed to the sun. Leaves undergoing rapid transpiration can be significantly cooler than the surrounding air. Transpiration is greatly affected by the species of plants that are in the soil, and it is strongly affected by the amount of light to which the plants are exposed. Water can be transpired freely by plants until a water deficit develops in the plant and its water-releasing cells (stomata) begin to close. Transpiration then continues at a much slower rate. Only a small portion of the water that plants absorb are retained in the plants.
- **Runoff** flows from a drainage basin or watershed that appears in surface streams. It generally consists of the flow that is unaffected by artificial diversions, storages, or other works that society might have on or in a stream channel.
- **Three basic locations of water storage occur in the planetary water cycle.** Water is stored in the atmosphere; water is stored on the surface of the earth, and water is stored in the ground.
- Water stored in the atmosphere can be moved relatively quickly from one part of the planet to another part of the planet. The type of storage that occurs on the land surface and under the ground largely depends on the geologic features related to the types of soil and the types of rocks present at the storage locations. Storage occurs as surface storage in oceans, lakes, reservoirs, and glaciers; underground storage occurs in the soil, in aquifers, and in the crevices of rock formations.



Water Resource

Fresh water





Water Resource

Water Use

Agriculture – 70%

Industry – 22%

Domestic – 8%

Water Storage Ground water – Aquifer

Helps!!

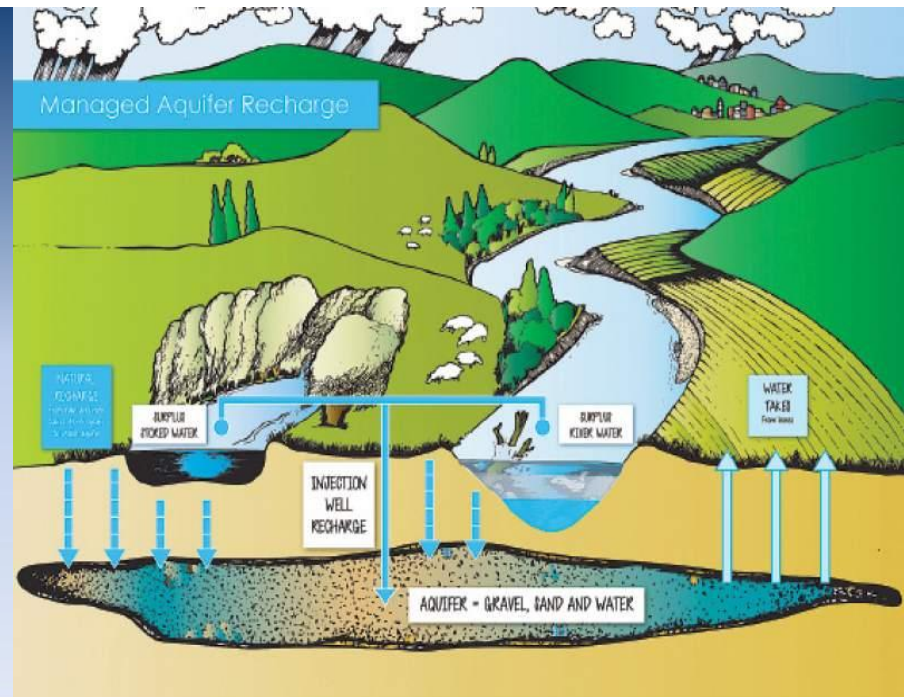
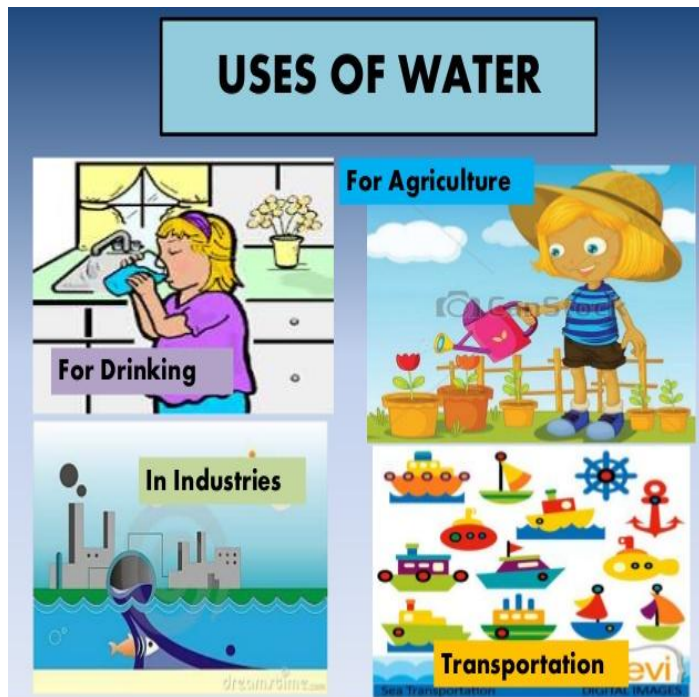
Potable water to drink

Reduce, Reuse, and Recycle
(3R)

Concern!!

Ground water depletion

Untreated ground water





Water Resource

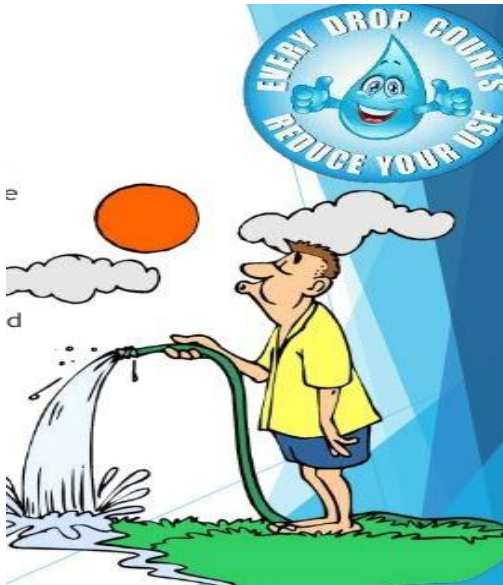
Water Storage – Surface water - Reservoir

Helps!!

- Energy Production
- Reduce Flood
- Irrigation
- Reclaimed water

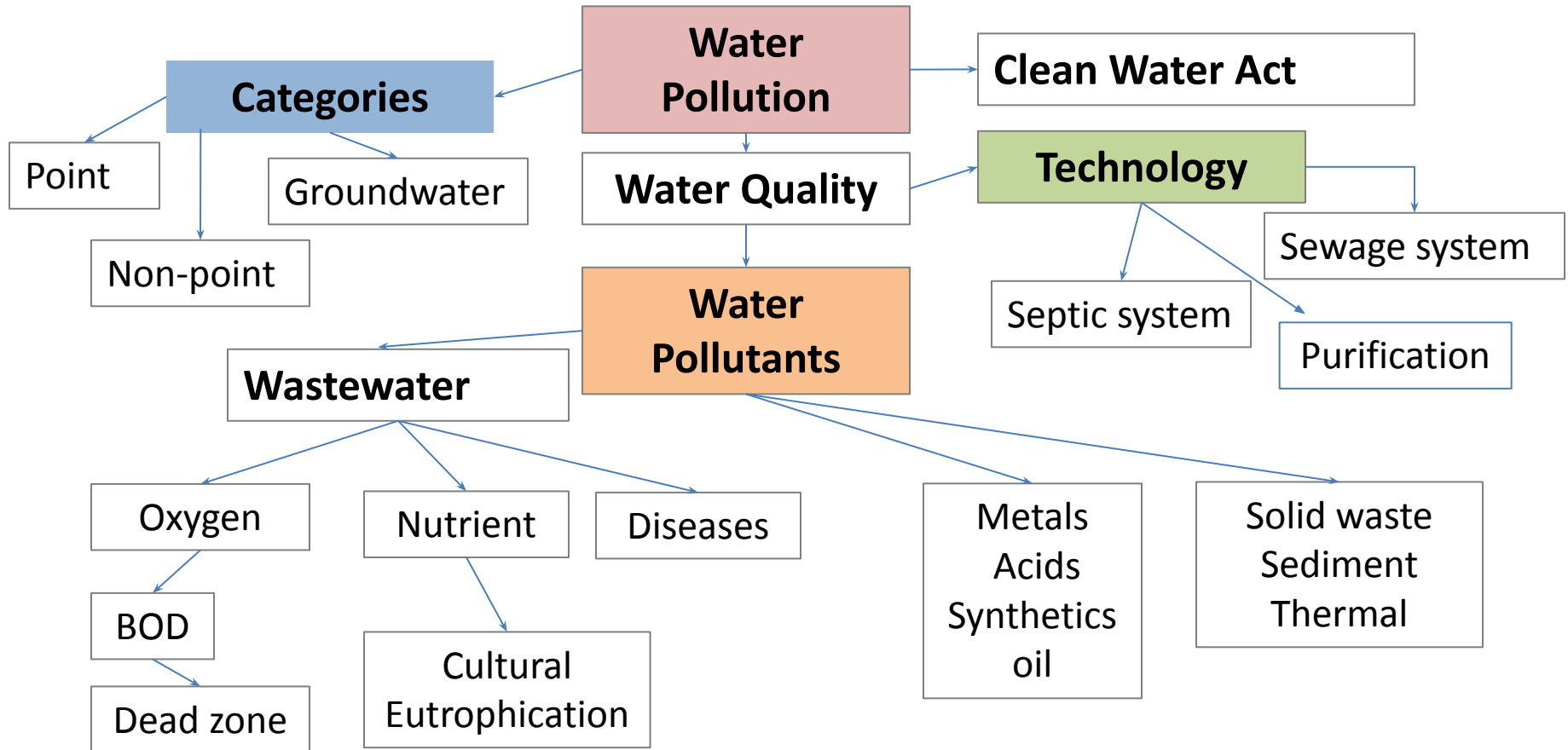
Concern!!

- Destruction
- Less Evaporation
- Wildlife hampered
- Eviction of people or community





Water Resource





Water Resource

Water Pollution

The contamination of surface water (streams, rivers , lakes, oceans) or groundwater with substances produced through human activities and natural phenomenon that negatively affect organisms.

Pollution Categories:

Point sources – distinct locations that pump waste or contaminants into a waterway. You can target (or point to).

Non-point sources – diffuse areas such as an entire farming region that pollutes a waterway. Hard to target where the source exactly is.

Groundwater pollution/depletion - long-term **water**-level/quality declines caused by sustained **groundwater** pumping a/o contamination



Water Resource

Categories



Point



Non-point



Groundwater



Water Resource

Water pollution

Causes

Sewage
Nutrients
Wastewater
Chemical waste
Radioactive waste
Oil pollution

Impacts

Death of aquatic (water) animals
Disruption of food-chains
Diseases
Destruction of ecosystems





Water Resource

Water quality

“**Water quality**” is a term used here to express the suitability of water to sustain various uses or processes. Any particular use will have certain requirements for the physical, chemical or biological characteristics of water (UNEP/WHO, 1996)

Water quality – parameters:

- **Physical parameters.** Colour, odour, turbidity, taste, temperature and electrical conductivity
- **Chemical parameters.** the number of carbonates, sulphates, chlorides, fluorides, nitrates, and metal ions
- **Biological parameters.** Quantity Existence of algae, fungi, viruses, protozoa and bacteria



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Water quality and dissolved oxygen (DO) content in parts per million (ppm) at 20°C

BOD Level in mg/liter	Water Quality
1 - 2	Very Good: There will not be much organic matter present in the water supply.
3 - 5	Fair: Moderately Clean
6 - 9	Poor: Somewhat Polluted - Usually indicates that organic matter present and microorganisms are decomposing that waste.
100 or more	Very Poor: Very Polluted - Contains organic matter.

- Only a few fish species can survive in the water less than 4ppm at 20°C



Water Resource

Water Pollutants

Types:

- (i) Sewage Pollutants (Domestic and Municipal Waste)
- (ii) Industrial Pollutants
- (iii) Agricultural Pollutants
- (iv) Radioactive and Thermal Pollutants

In the form of:

- ☐ Human Wastewater
- ☐ Chemical pollutants
- ☐ Non-chemical pollutants



Water Resource

Human Wastewater

Three reasons scientists are concerned about **wastewater**

1. Oxygen demanding waste like bacteria (**E.coli**) that put a large demand for oxygen in the water
2. Nutrients that are released from wastewater decomposition can make the water more fertile causing eutrophication
3. Wastewater can carry a wide variety of disease-causing organism.



Water Resource

Biochemical Oxygen Demand (BOD)

BOD – BOD provides a measure of the impact of waste (water) on the oxygen content in a receiving water body. Wastes (organic matter, "food") are broken down (decomposed, oxidized, metabolized) by microbial organisms (aerobic bacteria), frequently referred to as "bugs", who require oxygen for this monumental effort

- Lower BOD values indicate the water is less polluted and higher BOD values indicate it is more polluted by wastewater
- When bodies of water have high oxygen demand due to microbial decomposition, the amount of O_2 for other organisms can be very low



Water Resource

Wastewater



Decomposers



Dead zone



Water Resource

Eutrophication

Eutrophication is an abundance of fertility to a body of water. For instance, Algae bloom

- caused by an increase in nutrients (including nitrogen and phosphorus), such as fertilizers.
- can cause a rapid growth of algae which eventually dies, causing the microbes to increase the BOD.



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Cultural Eutrophication

Algae



Dead zone



Water Resource

diseases

Common diseases from Human Wastewater

- ✓ Cholera
- ✓ Typhoid fever
- ✓ Stomach flu
- ✓ Diarrhea
- ✓ Hepatitis

Facts!

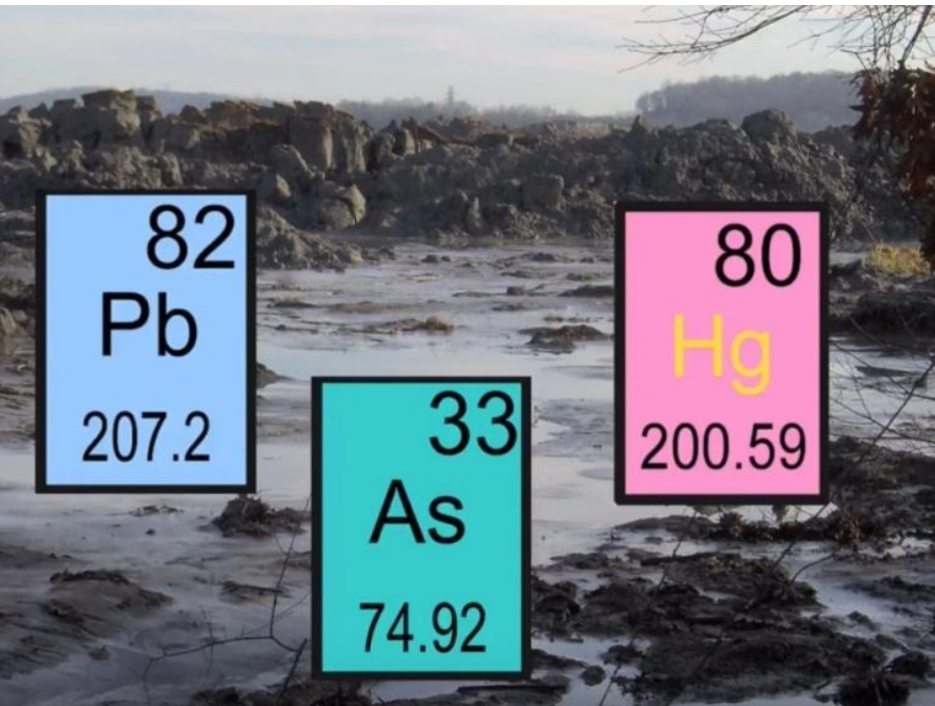
1/6 of the world's population does not have access to sufficient supplies of drinking water

½ of the diarrheal and malaria deaths could be prevented with safe drinking water, and proper sanitation and hygiene



Water Resource

Chemical Pollutants



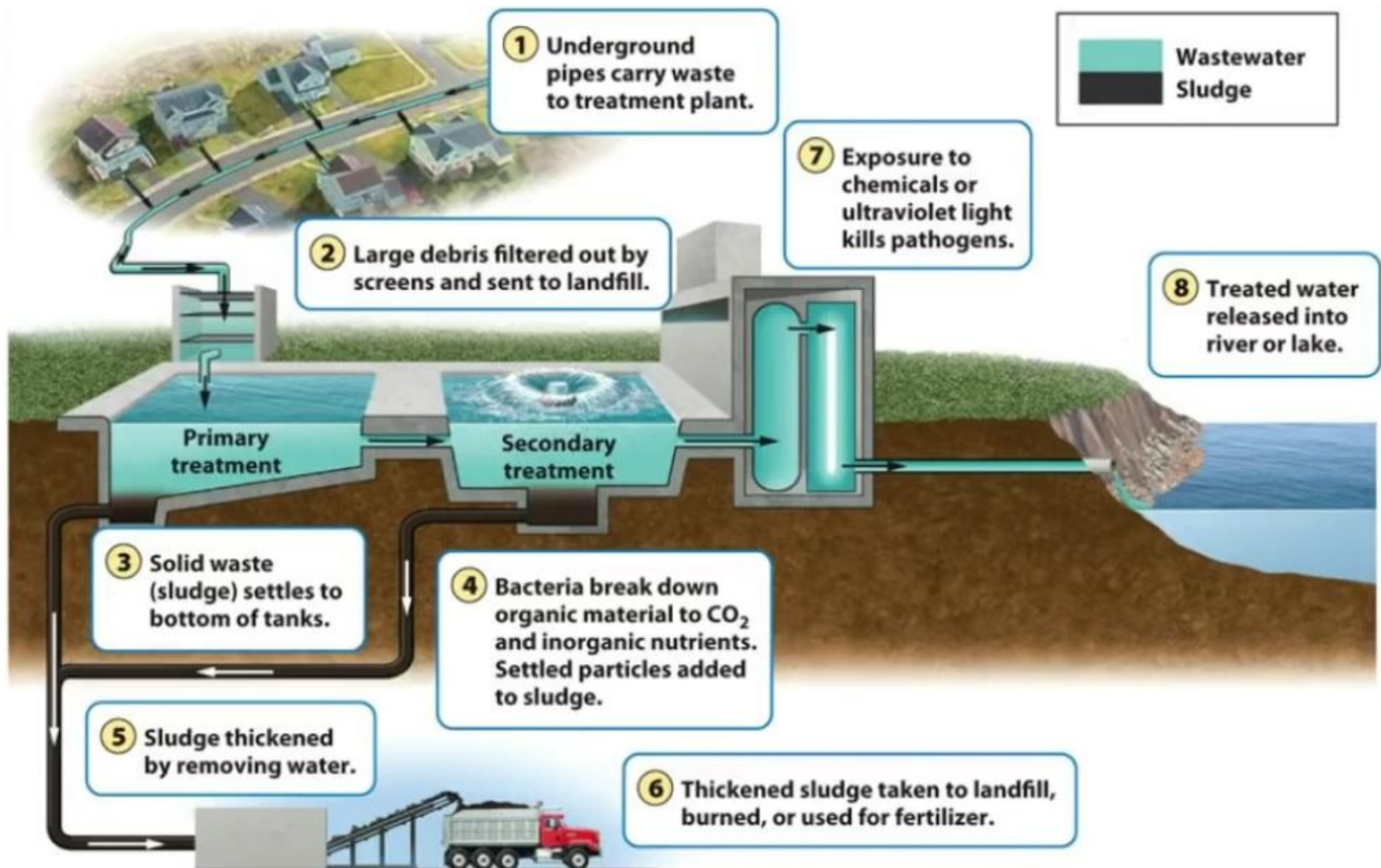
Non-chemical Pollutants





Water Resource

Sewage Treatment Plant





Water Resource

Treatments for Human and Animal Wastewater

- ▣ **Septic systems-** a large container that receives wastewater from the house. Anything that floats to the top is “scum.” Anything that sinks is “sludge.” The middle layer is clear water called “septage.” The septage contains bacteria, but also pathogenic organisms, N and P. The water flows through pipes and lawn called the leach field.



Figure 14.5
Environmental Science
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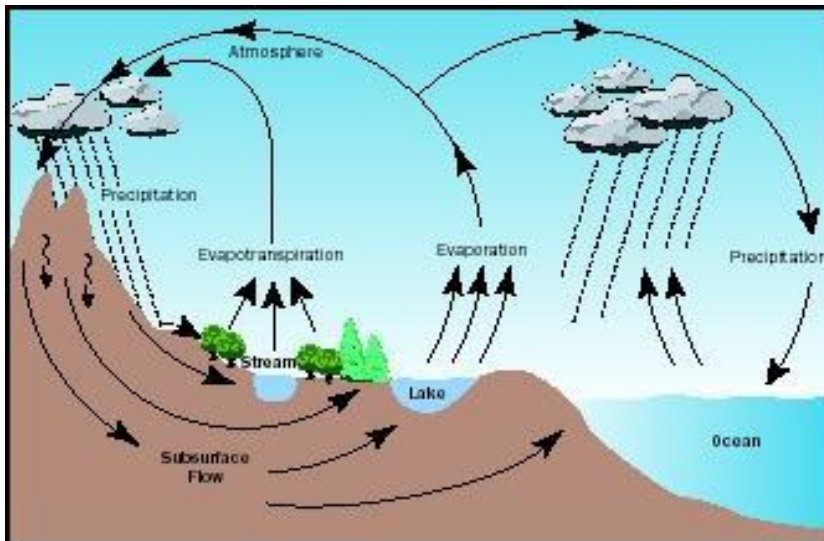
Pro - No electricity is used;
con - must be pumped out
every 5-10 years



Water Resource

Water Movement

Ground water Recharge –
Natural
Saturated flow
Irrigation
Potable water supply
Desalination



Water Conservation

- I. low-cost water through subsidies and incentives
- II. Water-saving Devices

Play your part, be water smart!

Wise Water Use



References

- https://gml.noaa.gov/outreach/lesson_plans/The%20Hydrologic%20Cycle.pdf
- <https://archive.nptel.ac.in/content/storage2/courses/105107129/module1/lecture1/lecture1.pdf>
- <http://waterandme.tamu.edu/WaterPollution/waterpollution.pdf>
- https://gcwgandhinagar.com/econtent/document/15876184390EVSAECC01_Water%20pollution.pdf